



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY UNA HIMACHAL PRADESH

An Institute of National Importance under MoI

Saloh, Una – 177 209

Website: www.iiitu.ac.in

School of Computing
CURRICULUM: IIITUGCSE23
Cycle Test - II
16, Apr.' 25
(02.00 PM-03.00PM)

Degree	B. Tech.	Branch	IT CSE
Semester	IV		
Subject Code & Name	CSC404: Computer Networks		
Time: 60 Minutes	Answer All Questions	Maximum: 20 Marks	

Sl. No.	Question	Marks
1.a	Explain three benefits of packet switching techniques.	1
1.b	A channel has a bit rate of 4 Kbps and one-way propagation delay of 20 msec. The channel uses stop and wait protocol. The transmission time of the acknowledgement frame is negligible. To get a channel efficiency of 50%, what should be the minimum frame size?	2
1.c	Construct the even parity Hamming code word for a data byte 1001101.	2
2.a	Explain five rules of ALOHA.	2
2.b	Consider two hosts X and Y connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2×10^8 m/sec. Host X sends a file of 50,000 bytes as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds respectively. Find the value of p and q?	2
2.c	What is back-off time in MAC?	2
3.a	Write the differences between Pure Aloha and Slotted Aloha.	1
3.b	Find the error bit location using even parity hamming code if the received code is 110001101.	2
3.c	A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then find the input bit-string.	2
4.a	Define "Store and Forward Method" in switching techniques.	1
4.b	Describe the connection-oriented switching method known as virtual circuit switching.	2
4.c	Station A needs to send a message consisting of 9 frames to station B using a sliding window (window size 3) and go back N error control strategy. All frames are ready and immediately available for transmission. What is the total number of frames that A will transmit to B if every fifth frame that A transmits is lost (but no ACKs from B are ever lost)?	2



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY UNA HIMACHAL PRADESH

An Institute of National Importance under MoE

Saloh, Una- 177 209

Website: www.iiitu.ac.in

School of Computing
CURRICULUM: IIITUGCSE23
Cycle Test - I
(20 Feb.'25, 02.00 PM-03.00PM)

Degree	B. Tech.	Branch	CSE
Semester	IV		
Subject Code & Name	CSC404: Computer Networks		
Time: 60 Minutes	Answer All Questions		Maximum: 20 Marks

Sl. No.	Question	Marks												
1.a	Find the maximum number of IP addresses that can be assigned to hosts on a local subnet that uses the 255.255.255.224 subnet mask.	1												
1.b	Explain the role of variable length subnet mask with suitable example.	2												
1.c	Draw the line coding of a bit stream "01001110" using unipolar, polar, bipolar encoding techniques.	2												
2.a	Differentiate between limited broadcast address and directed broadcast address with an example.	1												
2.b	Consider the following message $A=1010001101$. Find the cyclic redundancy check (CRC) for this message using the divisor polynomial is $x^5+x^4+x^2+1$ at the sender side. Also, it is assumed that the message A is found error at the 2 nd bit from MSB at the receiver side then find the CRC (remainder) at the receiver.	2												
2.c	A subnet has been assigned a subnet mask of 255.255.255.192. What is the maximum number of hosts that can belong to this subnet? Explain.	2												
3.a	How does "Packet" differ from "Segment" in OSI layers?	1												
3.b	What are the four effective characteristics of data communication? Explain.	2												
3.c	The routing table of a router is shown below: On which interface the router will forward the packets if the destination IP addresses are 128.75.43.16 and 192.12.17.10 respectively. Explain. <table><tr><th>Destination</th><th>Subnet Mask</th><th>Interface</th></tr><tr><td>128.75.43.0</td><td>255.255.255.0</td><td>Eth0</td></tr><tr><td>128.75.43.0</td><td>255.255.255.128</td><td>Eth1</td></tr><tr><td>192.12.17.5</td><td>255.255.255.255</td><td>Eth3</td></tr></table>	Destination	Subnet Mask	Interface	128.75.43.0	255.255.255.0	Eth0	128.75.43.0	255.255.255.128	Eth1	192.12.17.5	255.255.255.255	Eth3	2
Destination	Subnet Mask	Interface												
128.75.43.0	255.255.255.0	Eth0												
128.75.43.0	255.255.255.128	Eth1												
192.12.17.5	255.255.255.255	Eth3												
4.a	Explain the main responsibilities of OSI layers.	1												
4.b	Two devices A and B are configured as follows: A has IP address 203.197.2.53 and subnet mask 255.255.128.0. B has IP address 203.197.75.201 and subnet mask 255.255.192.0. A assumes B is on same network, but B assumes A is on a different network. Justify.	2												
4.c	Discuss various types of networks topologies in computer network. If $n=1098$ nodes, then find the number of cables and ports in mesh, and star topology.	2												



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY UNA HIMACHAL PRADESH

An Institute of National Importance under MoE

Saloh, Una – 177 209

Website: www.iiitu.ac.in

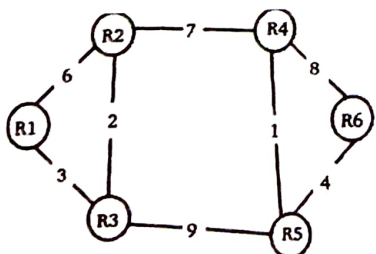
School of Computing CURRICULUM: IIITUGCSE23

End Semester Examination

08, May.'25

(9.00 AM-12.00 PM)

Degree	B. Tech.	Branch	CSE
Semester	IV		
Subject Code & Name	CSC404: Computer Networks		
Time: 180 Minutes	Answer All Questions	Maximum: 100 Marks	

Sl. No.	Question	Marks
1.a	Define Computer Networks? Discuss various types of networks topologies in Computer Network. Also discuss various advantages and disadvantages of each topology.	5
1.b	What is IP addressing? How it is classified? How subnet addressing is performed?	5
1.c	Explain the following: i. TCP ii. UDP	5
1.d	What is pure ALOHA and slotted ALOHA? Consider the delay of both at low load. Which one is less? Explain your answer.	5
2.a	In a block of addresses, we know the IP address of one host is 182.44.82.16/26. What are the first address (network address) and the last address in this block?	5
2.b	Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of slow start phase is 2 MSS and the threshold at the start of first transmission is 8 MSS. Assume that a time out occurs during the fifth transmission. Find the congestion window size at the end of tenth transmission.	5
2.c	Consider the effect of using slow start on a line with a 10 msec RTT and no congestion. The receiver window is 24 KB and the maximum segment size is 2 KB. How long does it take before the first full window can be sent?	5
2.d	If a Company require 60 hosts then what is the best possible subnet mask? Explain.	5
3.a	Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram: 	5

	All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data? Explain.	
3.b	If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?	5
3.c	Define the type of the following destination addresses: i. 4A:30:10:21:10:1A ii. 47:20:1B:2E:08:EE	5
3.d	In a class B subnet, we know the IP address of one host and the mask as given below: IP address: 125.134.112.66 Mask: 255.255.224.0 What will be the first address(Network address)?Explain.	5
4.a	Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link). What is the minimum number of bits that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.	5
4.b	There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot? Explain	5
4.c	The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. Find the message that should be transmitted.	5
4.d	Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, find the minimum number of bits required to represent the sequence number field.	5
5.a	Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?	5
5.b	An Internet Service Provider(ISP) has the following chunk of CIDR-based IP addresses available with it:245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. What is the valid allocation of addresses to A and B?	5
5.c	Host X has IP address 192.168.1.97 and is connected through two routers R1 and R2 to another host Y with IP address 192.168.1.80. Router R1 has IP addresses 192.168.1.135 and 192.168.1.110. R2 has IP addresses 192.168.1.67 and 192.168.1.155.The netmask used in the network is 255.255.255.224. Given the information above, how many distinct subnets are guaranteed to already exist in the network? Explain.	5
5.d	On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. Find the current window size at the sender.	5

$$\begin{array}{r}
 11001001011 \\
 1001 \\
 \hline
 1011001011 \\
 1001 \\
 \hline
 10001011 \\
 1001 \\
 \hline
 11011 \\
 1001 \\
 \hline
 1001
 \end{array}$$



AY 2023-24

School of Computing

CURRICULUM: IIITUGIT22

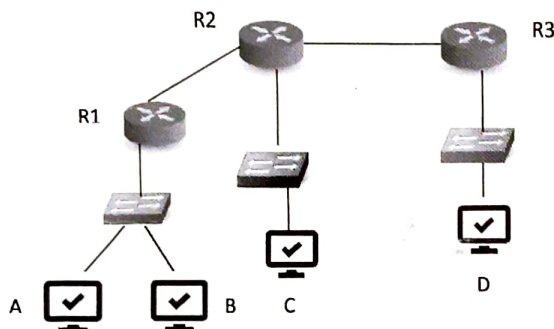
End-Sem. Lab. Examination

Series-B

01, May.'24: 09.00 am to 05.00 pm

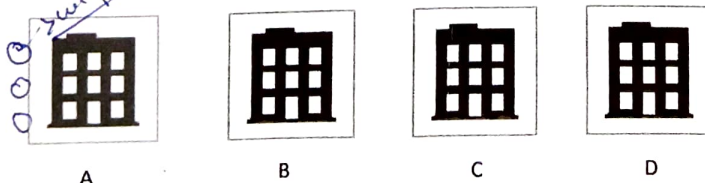
Degree	B. Tech.	Branch	IT
Semester	IV		
Subject Code & Name	ITC402: COMPUTER NETWORKS		
Answer All Questions			Maximum: 40 Marks

- Q1. An organization has implemented the campus network given in the Fig. 1. Design an IP scheme for the given network and implement the static routing in GNS3 so that all the computer systems can communicate with each other. (10+10)



The students are expected to use classful addressing concept for listing the IP scheme and demonstrate the concept of subnetting. The IP scheme should include IP address and Gateway for each of the network node shown in the diagram.

- Q2. The campus of an organization consists of the following buildings which should be connected with each other as a part of Campus Network. Design a logical network connectivity diagram for the campus and workout the Bill of Qty. for active and passive network components for the implementation of the same. It may be noted that Each building consists of 03 floors with each having 50 systems in a lab (dimension 20m * 30m) and within the labs systems should be connected through WiFi. Interbuilding distance is 200mtr. Implement the logical network connectivity in GNS3. (10+10)



Handwritten calculations and notes:

$25'8" \times 25'5" = 25'5"$

202

30